

2 Unit-II: Limits and Continuity

Introduction

Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad x \neq 2.$$

At $x = 2$, the expression is not defined because the denominator becomes zero. However, for $x \neq 2$,

$$x^2 - 4 = (x - 2)(x + 2),$$

and hence

$$f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2.$$

Now let us examine some values of $f(x)$ as x approaches 2.

x	$f(x)$
1.9	3.9
1.99	3.99
1.999	3.999
2.001	4.001
2.01	4.01
2.1	4.1

From the table, we observe that as x gets closer and closer to 2, the values of $f(x)$ get closer and closer to 4.

Therefore,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4.$$

Even though the original function is not defined at $x = 2$, its limit exists and equals 4.

This shows that the limit depends on the behavior of the function near the point and not necessarily on the value of the function at the point.

The idea of a *limit* describes the value that a function approaches when the independent variable approaches a particular point.

Let f be a real-valued function defined in a neighborhood of a , except possibly at a itself. We say that

$$\lim_{x \rightarrow a} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a , with $x \neq a$.

Notice that the value $f(a)$ is not important for the existence of the limit. The limit concerns the behavior of $f(x)$ near a .

Definition, Meaning and Geometrical Interpretation

Definition 2.1 (ε - δ Definition). We say that

$$\lim_{x \rightarrow a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$

Symbolically, $\forall \varepsilon > 0 \exists \delta > 0 \ni 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$

This is the rigorous definition of the limit of a function.

Meaning of ε :

The Greek letter

$$\varepsilon > 0$$

measures how close we want $f(x)$ to be to the proposed limit L .

The inequality

$$|f(x) - L| < \varepsilon$$

means that the distance between $f(x)$ and L is less than ε .

Equivalently,

$$L - \varepsilon < f(x) < L + \varepsilon.$$

Thus $f(x)$ lies inside the interval $(L - \varepsilon, L + \varepsilon)$.

This interval is called an ε -neighborhood of L .

Meaning of δ :

The number

$$\delta > 0$$

controls how close x must be to a .

The inequality

$$|x - a| < \delta$$

is equivalent to

$$a - \delta < x < a + \delta.$$

Because we exclude $x = a$, we write

$$0 < |x - a| < \delta.$$

Thus x lies in the deleted neighborhood

$$(a - \delta, a) \cup (a, a + \delta).$$

Geometrical Interpretation:

The definition

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

can be understood using two bands.

The condition

$$|x - a| < \delta$$

gives the vertical strip

$$a - \delta < x < a + \delta.$$

The condition

$$|f(x) - L| < \varepsilon$$

gives the horizontal strip

$$L - \varepsilon < y < L + \varepsilon.$$

Thus,

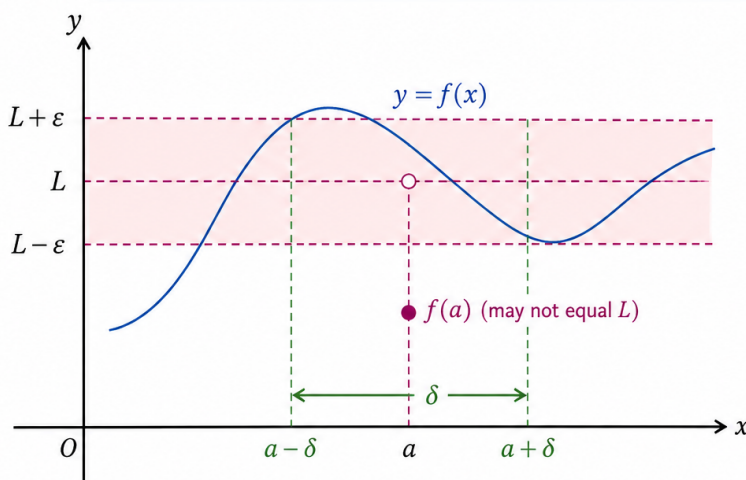
$$\lim_{x \rightarrow a} f(x) = L$$

means that, no matter how narrow a horizontal strip around L is chosen, we can find a sufficiently narrow vertical strip around a so that all points of the graph lying in that vertical strip, except possibly at $x = a$, lie inside the horizontal strip.

Geometrical Interpretation of a Limit

Let f be a function defined on an open interval containing a , except possibly at a .

We say $\lim_{x \rightarrow a} f(x) = L$ if and only if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a (but $x \neq a$).



Interpretation

- For any $\varepsilon > 0$, the horizontal band $L - \varepsilon < y < L + \varepsilon$ (the ε -neighbourhood of L) is shown by the shaded region.
- There exists $\delta > 0$ such that for all x with $0 < |x - a| < \delta$ (i.e., $a - \delta < x < a + \delta$, $x \neq a$), the points on the curve $y = f(x)$ lie inside the band $L - \varepsilon < y < L + \varepsilon$.
- The function value at $x = a$, $f(a)$, may be different from L (or $f(a)$ may not be defined).

Thus, $\lim_{x \rightarrow a} f(x) = L$ means that as x approaches a , the graph of $f(x)$ approaches the horizontal line $y = L$.

Why Do We Write $0 < |x - a|$?:

The condition

$$0 < |x - a|$$

means

$$x \neq a.$$

The value of the function at a itself does not determine the limit.

For instance,

$$f(x) = \begin{cases} x + 2, & x \neq 1, \\ 100, & x = 1. \end{cases}$$

Then

$$f(1) = 100,$$

but

$$\lim_{x \rightarrow 1} f(x) = 3.$$

Thus the limit depends on nearby values of the function, not necessarily on the value at the point.

Understanding the Logical Order:

The statement

$$\forall \varepsilon > 0, \quad \exists \delta > 0$$

is extremely important.

It means that first an arbitrary positive number ε is chosen. Then we must be able to find a suitable positive number δ .

Usually,

$$\delta = \delta(\varepsilon),$$

that is, δ may depend on ε , but δ must not depend on x .

The logical structure is

$$\boxed{\text{Given } \varepsilon > 0, \text{ find } \delta > 0 \text{ such that } |f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.}$$

The same ε - δ definition may be stated as follows:

Definition 2.2 (Neighborhood Form of the Definition). For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$f((a - \delta, a + \delta) \setminus \{a\}) \subseteq (L - \varepsilon, L + \varepsilon).$$

In words, every sufficiently small deleted neighborhood of a is mapped into the prescribed ε -neighborhood of L .

Examples Using ε - δ Definition

Problem 2.1. Prove that

$$\lim_{x \rightarrow 2} (3x + 1) = 7.$$

Solution

We want to show that for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - 2| < \delta$$

implies

$$|(3x + 1) - 7| < \varepsilon.$$

Now,

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|.$$

We want

$$3|x - 2| < \varepsilon.$$

This will happen if

$$|x - 2| < \frac{\varepsilon}{3}.$$

Therefore choose

$$\boxed{\delta = \frac{\varepsilon}{3}}.$$

Then, if

$$0 < |x - 2| < \delta,$$

we get

$$|(3x + 1) - 7| = 3|x - 2| < 3\delta = 3\left(\frac{\varepsilon}{3}\right) = \varepsilon.$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} (3x + 1) = 7}.$$

Problem 2.2. Prove that

$$\lim_{x \rightarrow 3} x^2 = 9.$$

Solution

We have

$$|x^2 - 9| = |x - 3||x + 3|.$$

We want

$$|x - 3||x + 3| < \varepsilon.$$

Suppose we require

$$|x - 3| < 1.$$

Then

$$2 < x < 4.$$

Therefore,

$$5 < x + 3 < 7,$$

so

$$|x + 3| < 7.$$

Thus,

$$|x^2 - 9| = |x - 3||x + 3| < 7|x - 3|.$$

To make this less than ε , require

$$|x - 3| < \frac{\varepsilon}{7}.$$

Therefore choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{7} \right\}.$$

Then

$$0 < |x - 3| < \delta$$

implies

$$|x^2 - 9| < 7|x - 3| < 7\delta \leq \varepsilon.$$

Hence,

$$\lim_{x \rightarrow 3} x^2 = 9.$$

Problem 2.3. Prove that

$$\lim_{x \rightarrow 1} (x^2 + x) = 2.$$

Solution

We have

$$|x^2 + x - 2| = |(x - 1)(x + 2)| = |x - 1||x + 2|.$$

Suppose

$$|x - 1| < 1.$$

Then

$$0 < x < 2.$$

Hence,

$$2 < x + 2 < 4,$$

so

$$|x + 2| < 4.$$

Therefore,

$$|x^2 + x - 2| < 4|x - 1|.$$

To make this less than ε , require

$$|x - 1| < \frac{\varepsilon}{4}.$$

Hence choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{4} \right\}.$$

Then

$$0 < |x - 1| < \delta$$

implies

$$|x^2 + x - 2| < 4|x - 1| < 4\delta \leq \varepsilon.$$

Therefore,

$$\lim_{x \rightarrow 1} (x^2 + x) = 2.$$

Problem 2.4. Prove that

$$\lim_{x \rightarrow a} (mx + c) = ma + c.$$

Solution

Consider

$$f(x) = mx + c.$$

We want to prove that

$$\lim_{x \rightarrow a} f(x) = ma + c.$$

Let

$$L = ma + c.$$

Then

$$\begin{aligned} |f(x) - L| &= |(mx + c) - (ma + c)| \\ &= |m(x - a)| \\ &= |m||x - a|. \end{aligned}$$

To have

$$|f(x) - L| < \varepsilon,$$

it is sufficient that

$$|x - a| < \frac{\varepsilon}{|m|}.$$

Thus, for $m \neq 0$, choose

$$\delta = \frac{\varepsilon}{|m|}.$$

Therefore,

$$\lim_{x \rightarrow a} (mx + c) = ma + c.$$

Uniqueness of Limit

Theorem 2.1. *If $\lim_{x \rightarrow a} f(x)$ exists, then it is unique.*

Proof

Suppose that $f(x)$ has two limits L and M as $x \rightarrow a$.

Thus, $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} f(x) = M$.

We shall prove that $L = M$.

If possible, assume that $L \neq M$. Then $|L - M| > 0$.

Choose $\varepsilon = \frac{1}{2}|L - M| > 0$.

Since $\lim_{x \rightarrow a} f(x) = L$, by the definition of limit, there exists a number $\delta_1 > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta_1$.

Similarly, since $\lim_{x \rightarrow a} f(x) = M$, there exists a number $\delta_2 > 0$ such that $|f(x) - M| < \varepsilon$ whenever $0 < |x - a| < \delta_2$.

Let $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have both $|f(x) - L| < \varepsilon$ and $|f(x) - M| < \varepsilon$.

Therefore, $|L - M| = |L - f(x) + f(x) - M| \leq |L - f(x)| + |f(x) - M|$ by the triangle inequality.

Hence, $|L - M| < \varepsilon + \varepsilon = 2\varepsilon$.

Using $\varepsilon = \frac{1}{2}|L - M|$, in the above, we obtain $|L - M| < 2 \cdot \frac{1}{2}|L - M| = |L - M|$, which is not possible.

Therefore, our assumption $L \neq M$ is false. Hence,

$$\boxed{L = M}.$$

Thus, if the limit of a function exists at a point, it is unique. □

Bounded Function

Definition 2.3 (Bounded Function). A real valued function f defined on a set A is said to be *bounded* on A if there exists a real number $M > 0$ such that

$$|f(x)| \leq M \text{ (i.e. } -M \leq f(x) \leq M \text{) for every } x \in A. \text{ (OR)}$$

Equivalently, f is bounded on A if there exist real numbers m and M such that

$$m \leq f(x) \leq M \text{ for every } x \in A. \text{ (OR)}$$

Simply, A real valued function $f(x)$ is bounded if its range is bounded.

Example 2.1. Consider $f(x) = \sin x$ on $\mathbb{R}$. Since $-1 \leq \sin x \leq 1$ for every $x \in \mathbb{R}$, we have $|\sin x| \leq 1$. Hence $f(x) = \sin x$ is bounded on $\mathbb{R}$.

Example 2.2. Consider $f(x) = \frac{1}{1+x^2}$ on $\mathbb{R}$. Since $x^2 \geq 0$, we have $1+x^2 \geq 1$. Therefore, $0 < \frac{1}{1+x^2} \leq 1$, which implies $-1 < 0 < \frac{1}{1+x^2} \leq 1$, Hence $\left| \frac{1}{1+x^2} \right| \leq 1$, and so f is bounded on $\mathbb{R}$.

Example 2.3. Consider $f(x) = x$ on the interval $[-2, 2]$. Then $-2 \leq x \leq 2$, and hence $|f(x)| = |x| \leq 2$. Therefore $f(x) = x$ is bounded on $[-2, 2]$. However, $f(x) = x$ is not bounded on $\mathbb{R}$, since its values can become arbitrarily large in both directions on $\mathbb{R}$.

Example 2.4. Consider $f(x) = \frac{1}{x}$ on the interval $(0, 1)$. As x approaches 0, $\frac{1}{x}$ becomes arbitrarily large. Hence there is no $M > 0$ such that $\left| \frac{1}{x} \right| \leq M$ for every $x \in (0, 1)$. Therefore, $f(x) = \frac{1}{x}$ is not bounded on $(0, 1)$.

Algebra of Limits

Lemma 2.1. If $\lim_{x \rightarrow a} f(x) = L$, then f is bounded in some deleted neighbourhood of a . Moreover, if $L \neq 0$, then $f(x) \neq 0$. That is, there exist constants $M > 0$ and $\delta > 0$ such that $|f(x)| \leq M$ whenever $0 < |x - a| < \delta$ and if $L \neq 0$, then $f(x) \neq 0$.

Proof

Since $\lim_{x \rightarrow a} f(x) = L$, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Now, by using the fact that $||c| - |d|| \leq |c - d|$, we have

$$||f(x)| - |L|| \leq |f(x) - L| < \varepsilon.$$

Therefore,

$$||f(x)| - |L|| < \varepsilon \quad \text{or} \quad -\varepsilon \leq |f(x)| - |L| < \varepsilon \quad \text{or} \quad |L| - \varepsilon \leq |f(x)| < |L| + \varepsilon. \quad (1)$$

Thus from right side inequality of (1), we have

$$|f(x)| < |L| + \varepsilon \implies f(x) \text{ is bounded, whenever } 0 < |x - a| < \delta.$$

Further if $L \neq 0$, take $\varepsilon = \frac{|L|}{2}$, therefore from (1), we have

$$|L| - \frac{|L|}{2} < |f(x)| < |L| + \frac{|L|}{2} \quad \text{or} \quad \frac{|L|}{2} < |f(x)| < \frac{3|L|}{2}, \text{ which implies that } f(x) \neq 0.$$

□

Theorem 2.2. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$.
(KUD: Jan-2026, Feb-2024, Mar-2022)

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \quad (\because |c + d| \leq |c| + |d|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow a} (f(x) + g(x)) = L + M}.$$

□

Theorem 2.3. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} (f(x) - g(x)) = L - M$.
(KUD: Feb-2023, Oct-2022)

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |(f(x) - g(x)) - (L - M)| &= |(f(x) - L) + (-g(x) + M)| \\ &\leq |f(x) - L| + |-g(x) + M| \quad (\because |c + d| \leq |c| + |d|) \\ &\leq |f(x) - L| + |g(x) - M| \quad (\because |-c| = |c|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow a} (f(x) - g(x)) = L - M.$$

□

Theorem 2.4. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} f(x)g(x) = LM$.

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}$ whenever $0 < |x - a| < \delta_1$.

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2(|L| + 1)}$ whenever $0 < |x - a| < \delta_2$.

Also, since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_3 > 0$ such that $|f(x)| < |L| + 1$ whenever $0 < |x - a| < \delta_3$. (see, Lemma 1 for $\varepsilon = 1$)

Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |f(x)g(x) - LM| &= |f(x)g(x) - f(x)M + f(x)M - LM| \\ &= |f(x)(g(x) - M) + M(f(x) - L)| \\ &\leq |f(x)| |g(x) - M| + |M| |f(x) - L| \\ &< (|L| + 1) \frac{\varepsilon}{2(|L| + 1)} + |M| \frac{\varepsilon}{2(|M| + 1)} \\ &< \frac{\varepsilon}{2} + 1 \cdot \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence,

$$\lim_{x \rightarrow a} f(x)g(x) = LM.$$

□

Theorem 2.5. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, where $M \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon|M|}{4}$ whenever

$$0 < |x - a| < \delta_1.$$

Since $\lim_{x \rightarrow a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon |M|^2}{4(|L| + 1)}$ whenever $0 < |x - a| < \delta_2$.

Also, since $\lim_{x \rightarrow a} g(x) = M$ and $M \neq 0$, there exists $\delta_3 > 0$ such that $\frac{|M|}{2} < |g(x)|$ whenever $0 < |x - a| < \delta_3$. (see, Lemma 1)

Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{L}{M} \right| &= \left| \frac{Mf(x) - Lg(x)}{Mg(x)} \right| = \left| \frac{Mf(x) - LM + LM - Lg(x)}{Mg(x)} \right| \\ &= \left| \frac{M(f(x) - L) + L(M - g(x))}{Mg(x)} \right| \\ &\leq \frac{|M| |f(x) - L| + |L| |g(x) - M|}{|M| |g(x)|}. \end{aligned}$$

Since $|g(x)| > \frac{|M|}{2} \implies \frac{1}{|g(x)|} < \frac{2}{|M|}$, we obtain

$$\begin{aligned} \left| \frac{f(x)}{g(x)} - \frac{L}{M} \right| &< \frac{2}{|M|^2} (|M| |f(x) - L| + |L| |g(x) - M|) \\ &< \frac{2}{|M|^2} \left(|M| \frac{\varepsilon |M|}{4} + |L| \frac{\varepsilon |M|^2}{4(|L| + 1)} \right) \\ &= \frac{\varepsilon}{2} + \frac{|L|}{(|L| + 1)} \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + 1 \cdot \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Therefore,

$$\boxed{\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}}.$$

□

Theorem 2.6. If $\lim_{x \rightarrow a} f(x) = L$, then $\lim_{x \rightarrow a} |f(x)| = |L|$.

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x \rightarrow a} f(x) = L$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Therefore, whenever $0 < |x - a| < \delta$, we have $||f(x)| - |L|| \leq |f(x) - L| < \varepsilon$.

Hence,

$$\boxed{\lim_{x \rightarrow a} |f(x)| = |L|}.$$

□

Note (Converse:). The converse of the above theorem is *not true*.

That is, the statement $\lim_{x \rightarrow a} |f(x)| = |L|$ does not necessarily imply that $\lim_{x \rightarrow a} f(x) = L$.

Example 2.5 (Counterexample).

Define $f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$

We have $|f(x)| = 1$ for every $x \neq 0$. Therefore, $\lim_{x \rightarrow 0} |f(x)| = 1$.

However, $\lim_{x \rightarrow 0^+} f(x) = 1$ and $\lim_{x \rightarrow 0^-} f(x) = -1$.

Since, the right-hand and left-hand limits are different, $\lim_{x \rightarrow 0} f(x)$ does not exist.

Hence, the converse of the above theorem is false.

$\therefore \boxed{\lim_{x \rightarrow a} f(x) = L \implies \lim_{x \rightarrow a} |f(x)| = |L|}, \text{ but } \boxed{\lim_{x \rightarrow a} |f(x)| = |L| \not\Rightarrow \lim_{x \rightarrow a} f(x) = L}.$